

EFMW ZOO OUT-OF-SAMPLE EXPANSION PROTOCOL

ZOOS-OOS-1

Frozen-Core Validation and Prospective Animal Development Method

Status: Proposed canonical methodology

Core corpus: EFMW Zoo Animals #1–46

Extension corpus: Animals #47 onward

Primary principle: The original 46 are frozen. New animals are out-of-sample.

Abstract

The EFMW Zoo consists of 46 published experimental frameworks, each represented by an animal corresponding to a particular observational, diagnostic, operational, adversarial, temporal, relational, or systemic habitat.

Once this 46-animal architecture exists, continued development creates a methodological problem.

If Animals #47 onward are designed using the same benchmark environments, failure cases, thresholds, scoring rules, and observations that shaped Animals #1–46—and are subsequently folded back into those benchmarks—the Zoo risks becoming progressively optimized to its own evaluation environment.

The **EFMW Zoo Out-of-Sample Expansion Protocol (ZOOS-OOS-1)** prevents this.

The protocol freezes Animals #1–46 as the **Core Zoo** and treats all subsequently developed animals as **prospective out-of-sample extensions**. New animals may inherit general design principles from the Core Zoo, but they may not be used retrospectively to modify the original benchmark definitions or improve the apparent historical performance of the original 46.

The central scientific question becomes:

Does the methodology that produced the original EFMW Zoo generalize to previously unseen problem habitats?

This converts Zoo expansion itself into a prospective experiment.

1. The Frozen Core

The following set is designated:

EFMW Zoo Core v1.0 — Animals #1–46

The membership of this set is frozen.

The identities, intended habitats, principal metrics, benchmark definitions, baseline comparisons, and falsification conditions associated with these animals constitute the historical development corpus.

The Core Zoo may be:

- documented;
- debugged;
- reproduced;
- independently implemented;
- tested on additional datasets;
- benchmarked against stronger baselines;
- subjected to additional ablations.

It may **not**, for purposes of the original benchmark record, be retrospectively redesigned in response to the behavior of later animals.

This distinction is crucial.

Software maintenance is permitted.

Experimental history rewriting is not.

2. Why the Freeze Exists

Suppose Animal #51 reveals that a particular temporal statistic performs exceptionally well.

If that statistic is subsequently inserted into Animals #1–46 and their original benchmarks are rerun, the resulting performance is no longer genuinely prospective.

Information discovered after the original framework existed has leaked backward into its evaluation.

This creates several familiar statistical problems:

benchmark contamination, researcher degrees of freedom, multiple-hypothesis search, post-hoc optimization, selection bias, and eventually **overfitting to the research program itself**.

The Zoo therefore treats historical information as having a direction.

Core development→freeze→OOS development→prospective evaluation

Information may move forward.

For primary validation, it must not move backward.

3. Two Experimental Populations

From the freeze point forward, the Zoo contains two formally distinct populations.

Population A — Core Zoo

Animals:

$ZC=\{Z1,Z2,\dots,Z46\}$

These constitute the **development set**.

They establish the initial vocabulary of Zoo operations.

Population B — OOS Zoo

Animals:

$ZO=\{Z47,Z48,Z49,\dots\}$

These constitute the **prospective extension set**.

Their scientific purpose is not simply to make the Zoo larger.

Their purpose is to test whether the method that generated the Core Zoo continues to produce useful frameworks in habitats that were not responsible for shaping the Core.

4. The Information Firewall

The central mechanism of ZOOS-OOS-1 is an **information firewall**.

A new animal may know:

- the architecture of the existing Zoo;
- the general EFMW methodology;
- the existence of previous animals;
- established scientific baselines;
- general lessons about experimental design;
- failures discovered during Core development.

A new animal should **not be optimized against the original Zoo benchmark outcomes**.

In particular, the OOS development process should avoid choosing:

- equations because they improve an old Zoo score;
- thresholds because they rescue a failed Core experiment;
- weights because they maximize historical Zoo performance;
- features because they distinguish previously known benchmark labels;
- habitats solely because existing Zoo benchmarks happen to favor them.

The distinction is:

Methodological inheritance is allowed. Benchmark-answer inheritance is not.

5. The Animal Admission Test

A proposed OOS animal must first answer:

What problem does this animal solve that is not adequately represented by Animals #1–46?

A candidate should not receive a new animal identity merely because it uses different terminology.

It must establish a distinct **experimental habitat**.

Candidate #47 must therefore demonstrate at least one substantial novelty dimension, such as:

- a new observational structure;
- a new temporal regime;

- a new causal structure;
- a new form of uncertainty;
- a new interaction topology;
- a new failure mechanism;
- a new scale;
- a new intervention problem;
- a new human/system relationship;
- a new form of emergence.

If an existing animal already addresses the problem, the proposal should ordinarily become an extension or experiment of that animal instead.

This is the Zoo's first defense against taxonomic inflation.

6. Pre-registration of a New Animal

Before primary benchmark execution, every OOS animal should receive a frozen specification.

At minimum it contains:

Animal identity

Name and acronym.

Habitat

The class of systems in which an advantage is predicted.

Target phenomenon

Exactly what structure the animal attempts to detect, predict, explain, or control.

Inputs

What information it is permitted to observe.

Outputs

What it produces.

Primary metric

The principal quantity on which success or failure will be judged.

Baselines

The conventional methods against which it will be compared.

Compute budget

Where meaningful, competing methods receive comparable computational resources.

Information budget

The animal and its baselines receive equivalent access to relevant information.

Ablations

Which components will be removed to determine whether the proposed mechanism actually matters.

Null hypothesis

What result means that the animal added nothing.

Success criterion

Defined before results are observed.

Failure criterion

Also defined before results are observed.

The specification is then frozen.

7. The Habitat Hypothesis

Every animal should state a habitat hypothesis of the form:

Animal X is expected to outperform appropriate conventional baselines primarily when conditions H are present.

This is stronger than claiming:

Animal X works.

For example, an animal might predict increasing advantage as:

- temporal depth increases;
- partial observability increases;
- feedback becomes stronger;
- relational dependency increases;
- contradiction becomes distributed;
- state becomes more persistent.

It might simultaneously predict that its advantage disappears when:

- the problem becomes memoryless;
- observations become complete;
- relationships become irrelevant;
- noise exceeds recoverable structure.

That second prediction is important.

A useful theory should predict **where it should lose**.

8. OOS Benchmark Construction

The primary benchmark for a new animal must not simply be another sample from the exact environment used to design it.

At minimum, evaluation should distinguish:

Development environment

Used for implementation and debugging.

Validation environment

Used for bounded model selection.

Held-out environment

Not used for parameter selection.

OOS environment

Structurally different from the development ecology and reserved for prospective testing.

Whenever feasible, the OOS environment should vary dimensions such as:

- topology;
- noise;
- scale;
- temporal horizon;
- perturbation family;
- observation structure;
- parameter distribution;
- failure mechanism.

The strongest test is not:

Can the animal recognize another example of what created it?

It is:

Can the animal retain useful behavior when the surface characteristics of the problem change?

9. Baseline Equality

A new animal must compete against serious conventional alternatives.

Weak baselines artificially manufacture novelty.

Depending on the habitat, these might include:

- state-space models;
- Kalman methods;
- CUSUM;
- EWMA;
- change-point detection;
- graph algorithms;
- Bayesian methods;
- conventional anomaly detection;
- causal inference;
- survival analysis;
- time-series forecasting;

- control-theoretic monitors;
- modern machine-learning approaches;
- appropriately tuned ensembles.

The baseline should be something a competent practitioner would actually use.

10. Matched Information

A particularly important rule is:

$$I_{\text{Zoo}} \approx I_{\text{baseline}}$$

where I represents available information.

A Zoo animal must not receive historical context, relational variables, privileged labels, or future-derived features denied to its comparison method unless that asymmetry is itself the object of the experiment.

Otherwise the experiment measures information advantage rather than methodological advantage.

11. Matched False-Alarm Rates

For warning and detection animals, lead-time comparisons must normally occur at comparable false-alarm rates.

A detector that fires constantly will inevitably detect failures earlier.

Therefore:

$$\text{FPR}_{\text{Zoo}} \approx \text{FPR}_{\text{baseline}}$$

before lead-time advantage is interpreted.

This principle applies equally to OOS animals.

12. Ablation Before Interpretation

If a new animal succeeds, the next question is not immediately:

What can we deploy this on?

It is:

Why did it succeed?

Suppose an animal contains components:

$A+B+C+D$

Run:

$A+B+C$ $A+B+D$ $A+C+D$ $B+C+D$

and, where practical, individual components and relevant interactions.

If removing C destroys the effect while removing the others does not, the mechanism has become substantially more informative.

If every component can be removed without consequence, the proposed architecture is not supported.

13. Negative Results Are Zoo Results

A failed animal is not automatically removed.

An OOS animal can produce several scientifically meaningful outcomes:

PASS — preregistered advantage survives.

PARTIAL — effect exists only within a narrower habitat.

REDUNDANT — another method explains the effect equally well.

NULL — no reliable advantage is detected.

FAILED MECHANISM — performance occurs, but the proposed operation does not cause it.

FAILED GENERALIZATION — development performance disappears OOS.

These classifications should remain in the record.

A Zoo consisting only of winners would provide very little evidence that its falsification machinery works.

14. No Retroactive Rescue

This is the strongest ZOOS-OOS rule.

If Animal #52 fails its preregistered test, its primary result remains a failure.

It may subsequently be revised.

But the revision becomes:

52.1

or another explicitly identified successor experiment.

The original result remains preserved.

Likewise, a new animal cannot retroactively alter the primary historical result of Animal #17.

New knowledge can motivate a new experiment.

It cannot rewrite the old one.

15. Cross-Animal Composition

After independent evaluation, OOS animals may be combined with Core animals.

For example:

Z7+Z31+Z48

might form a composite monitor.

But the composite constitutes a **new experimental object**.

Its performance cannot be attributed automatically to its constituent animals.

The composite must receive:

- its own specification;
- its own baselines;
- its own ablations;
- its own OOS evaluation.

This prevents combinatorial Zoo systems from escaping falsification.

16. Generational Structure

The Zoo can consequently be organized by experimental generation.

Generation 0 — Core

G0=Z1..46

Frozen historical development corpus.

Generation 1 — First OOS cohort

G1=Z47..n

Developed after the Core freeze.

Generation 2

Developed only after Generation 1 evaluation is frozen.

And so forth.

Each generation therefore becomes a prospective test of knowledge learned from the previous generation.

This produces a particularly useful structure:

G0→G1→G2→G3

Rather than one endlessly modified benchmark.

17. Cross-Generation Testing

Once several generations exist, a stronger experiment becomes possible.

An animal developed in G1 can be evaluated against environments introduced in G2 or G3.

Thus the Zoo can construct a matrix:

M_{ij} =performance of generation i on environment generation j

This makes it possible to distinguish:

- genuine generalization;
- generational overfitting;
- habitat-specific specialization;
- methodological improvement.

18. The Meta-Hypothesis

ZOOS-OOS-1 ultimately tests something larger than individual animals.

The meta-hypothesis is:

The EFMW Zoo methodology identifies relational, temporal, recursive, causal, contextual, and systemic structures that retain useful predictive or explanatory value outside the benchmark ecology in which the methodology originated.

That claim cannot be established by Animals #1–46 alone.

Those animals helped create the methodology.

Animals #47 onward provide the opportunity to test it.

19. The Strongest Possible Result

The strongest evidence would not be:

We designed 100 animals and most performed well.

It would be something closer to:

We froze 46 frameworks and their methodology. We subsequently specified new frameworks prospectively, tested them against strong baselines in previously unseen habitats, preserved negative results, and observed reproducible performance patterns predicted before evaluation.

That would represent a qualitatively stronger evidentiary standard.

20. Repository Convention

Every post-Core animal should contain a machine-readable declaration such as:

```
EFMW_ZOO_PROTOCOL = ZOOS-OOS-1
CORE_VERSION = 1.0
CORE_RANGE = 1-46
ANIMAL_NUMBER = 47
GENERATION = 1
DEVELOPMENT_STATUS = OOS
CORE_BENCHMARK_TUNING = PROHIBITED
PRIMARY_SPECIFICATION = FROZEN_BEFORE_TEST
NEGATIVE_RESULTS = PRESERVED
```

Its repository should additionally contain:

```
README
THEORY
SPECIFICATION
PREREGISTRATION
BASELINES
BENCHMARK
ABLATIONS
RESULTS
FALSIFICATION
CHANGELOG
```

The **CHANGELOG** becomes especially important because it establishes when information entered the experimental process.

21. Canonical Freeze Statement

The following statement can be placed in the Zoo master repository and the future OOS repositories:

EFMW Zoo Core Freeze

Animals #1–46 constitute EFMW Zoo Core v1.0. They are treated as the historical development corpus for subsequent Zoo research. Animals developed after the Core freeze are designated out-of-sample extensions. Results from those extensions shall not be used retrospectively to alter the primary benchmark definitions, thresholds, model-selection decisions, or historical evaluation claims of the Core Zoo. New findings may motivate explicitly versioned follow-up experiments, but original results remain preserved. The purpose of this separation is to test whether the Zoo methodology generalizes beyond the experimental ecology that produced the original 46 frameworks.

22. The Rule in One Line

The entire protocol can ultimately be reduced to:

Freeze what taught you; test what you learned on something that did not.

That is the methodological significance of keeping Animal #47 outside the original 46-animal benchmark ecology.

It turns the next Zoo expansion from **more framework construction** into a prospective test of whether the framework-construction method itself generalizes.